

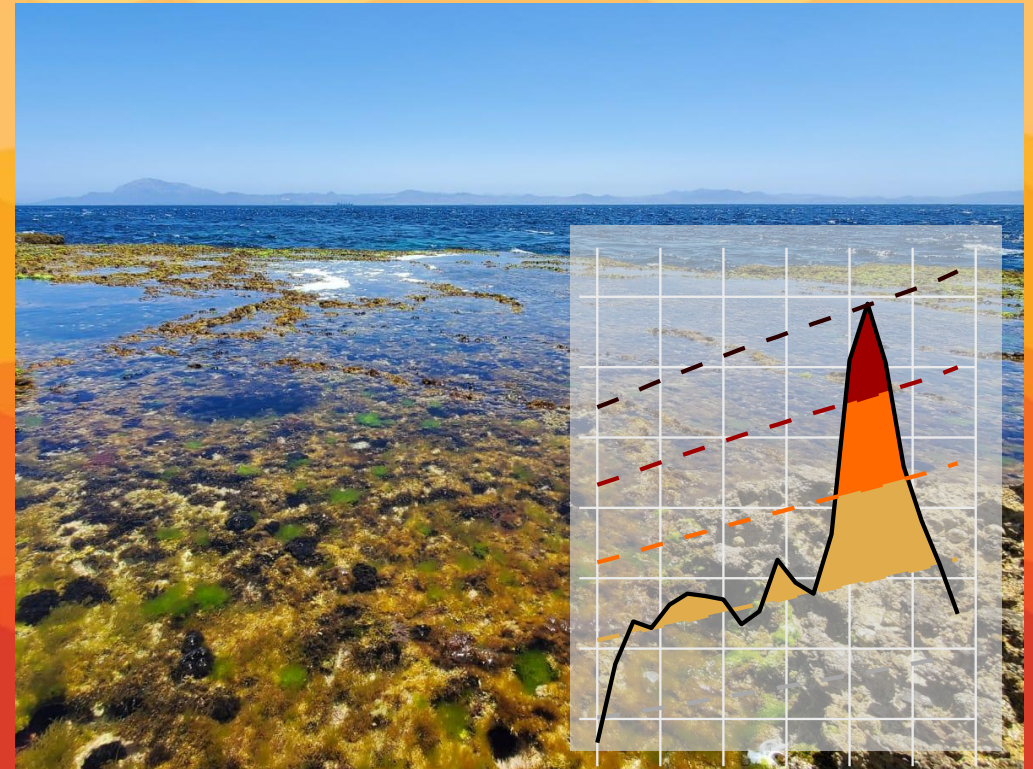


WALKING THE THERMAL LANDSCAPE: *TIME EXPLICIT THERMAL TOLERANCE MODELS IMPROVE OUR UNDERSTANDING OF MARINE HEATWAVES' BIOLOGICAL EFFECTS*

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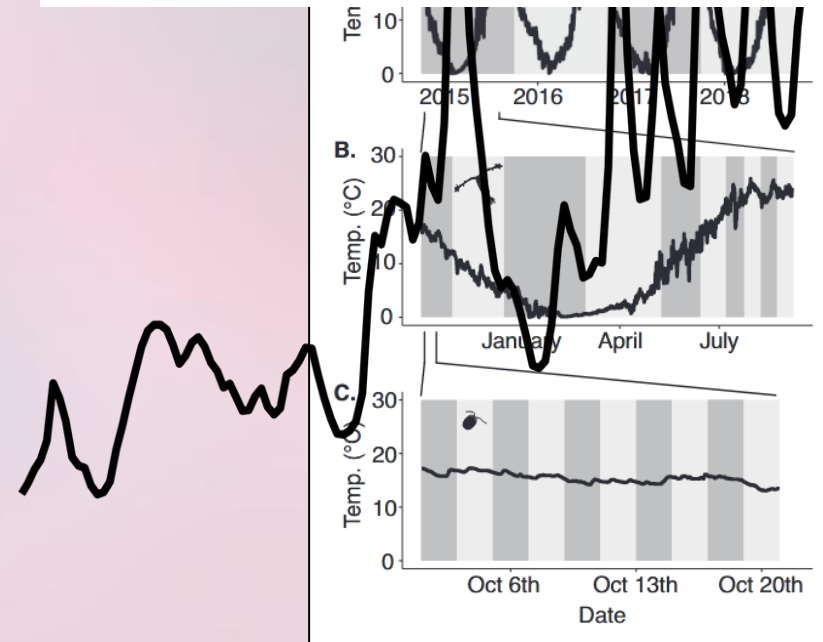
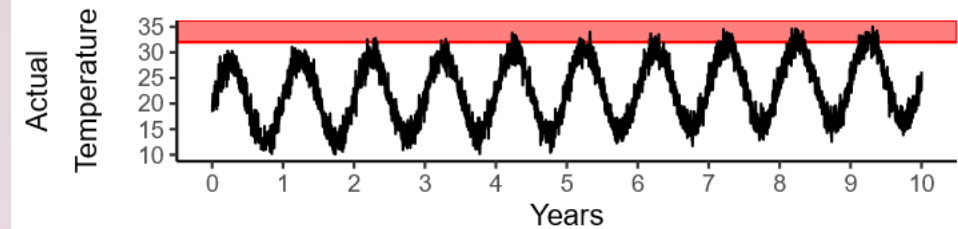
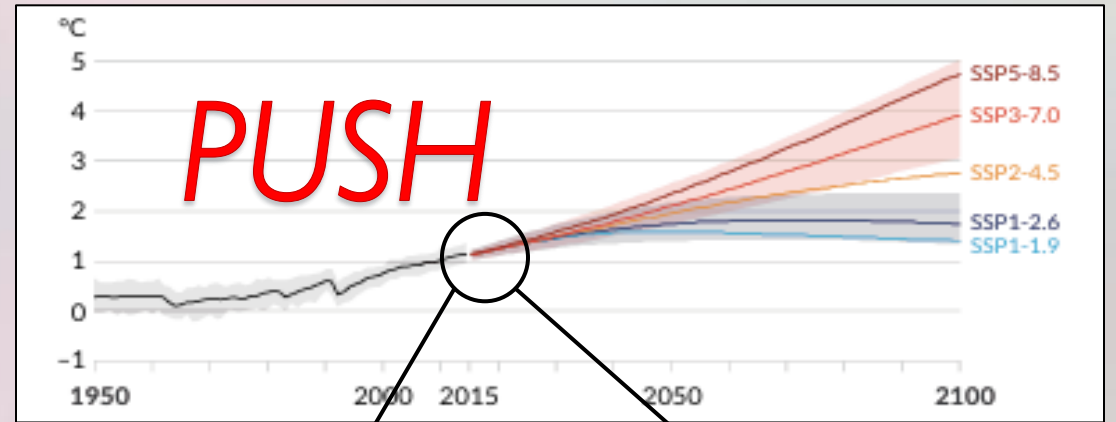
Bigelow Laboratory for Ocean Sciences
11 December 2025



PREDICTING POPULATIONS UNDER A VARIABLE FUTURE

- The rise of extreme heatwaves over the push of climate change challenges predictions of the future.
- We see extreme temperature events happening on much quicker time scales — pulses occurring minutes to months (Oliver et al. 2019).
- This is the scale at which many animals operate.

IPCC RCP Global mean temperature projections



Sasaki et al. 2025,
Nature
Communications

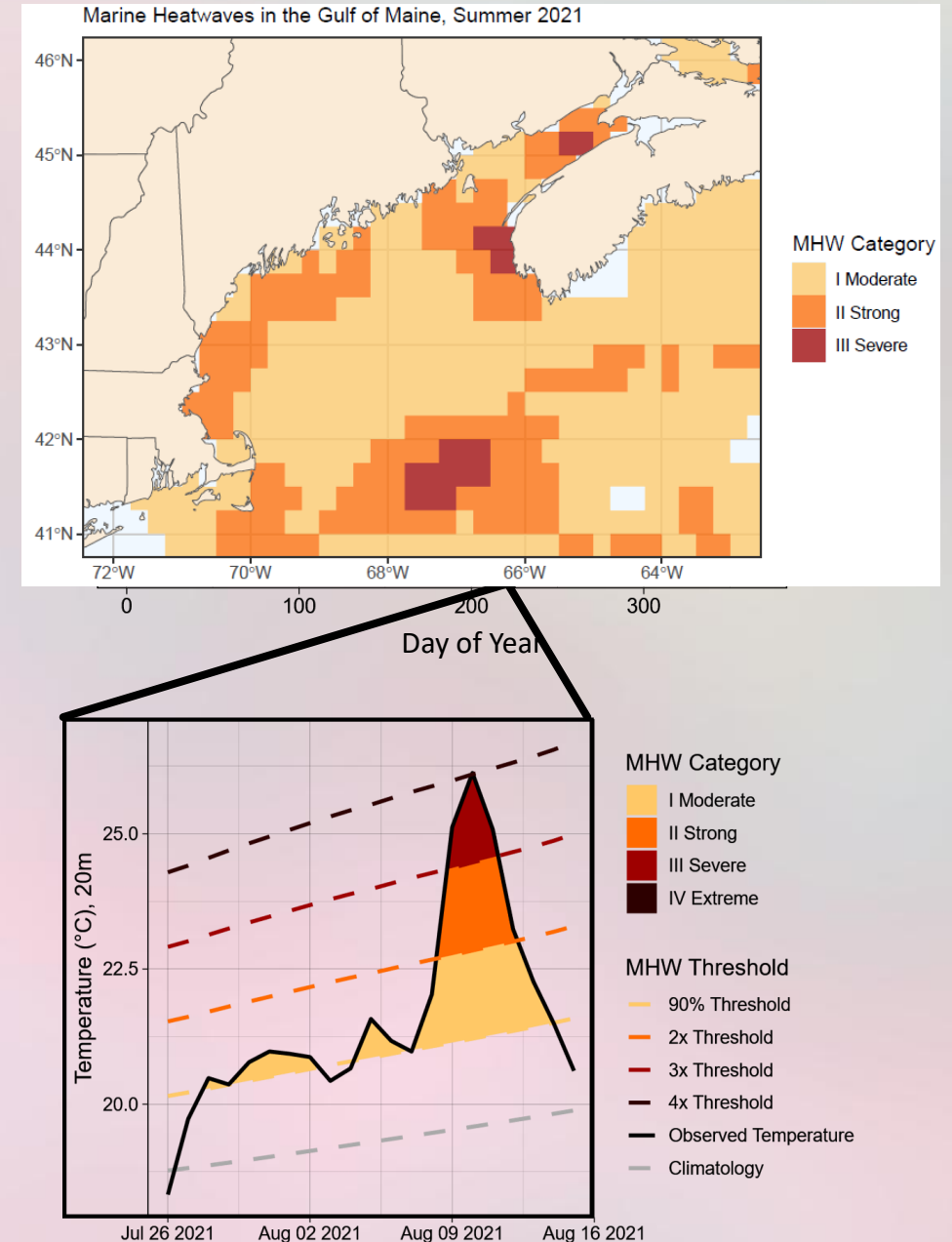
TRANSLATING HEATWAVES: FROM STATISTICS TO BIOLOGY

Marine Heatwaves (MHWs)

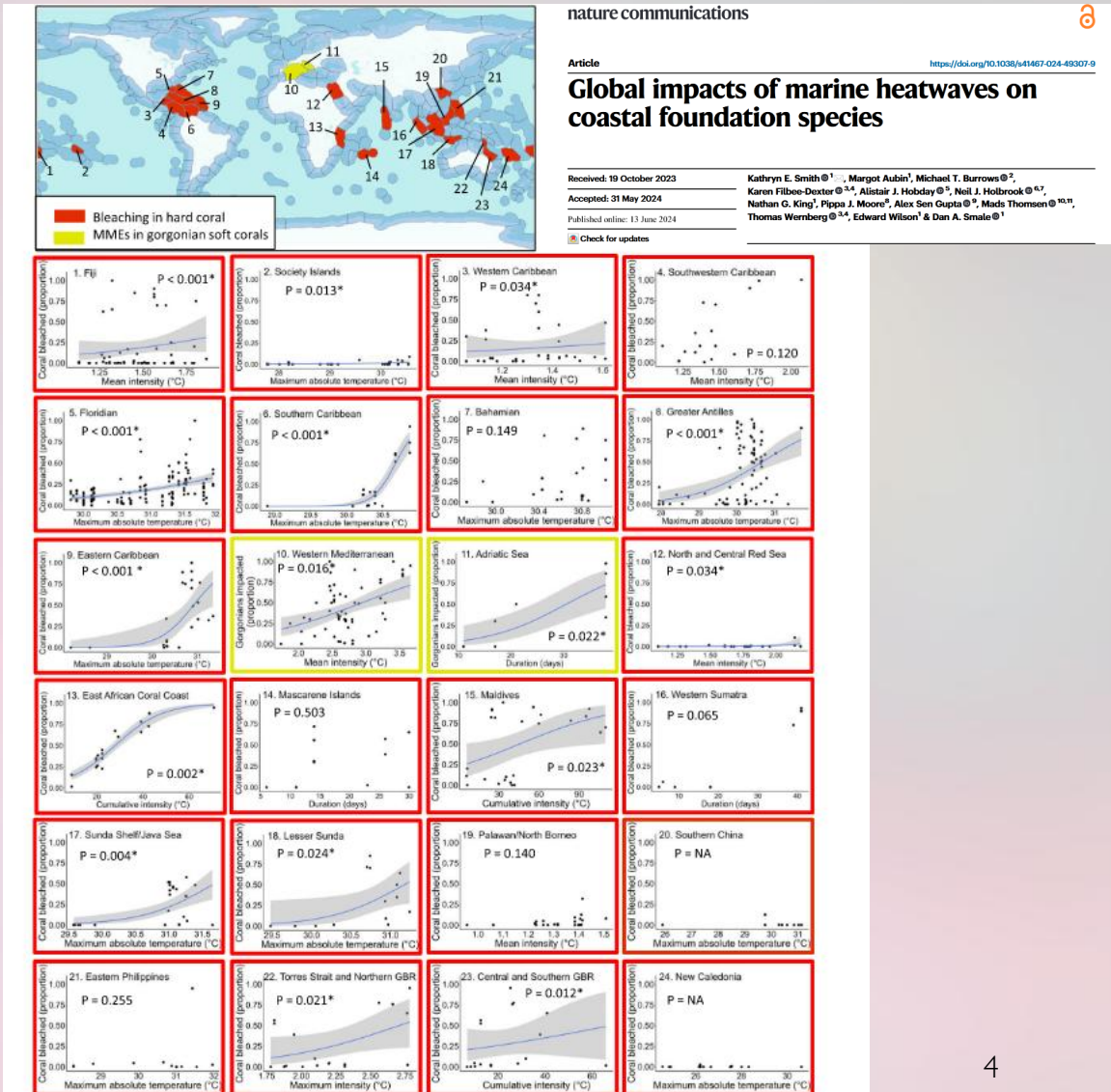
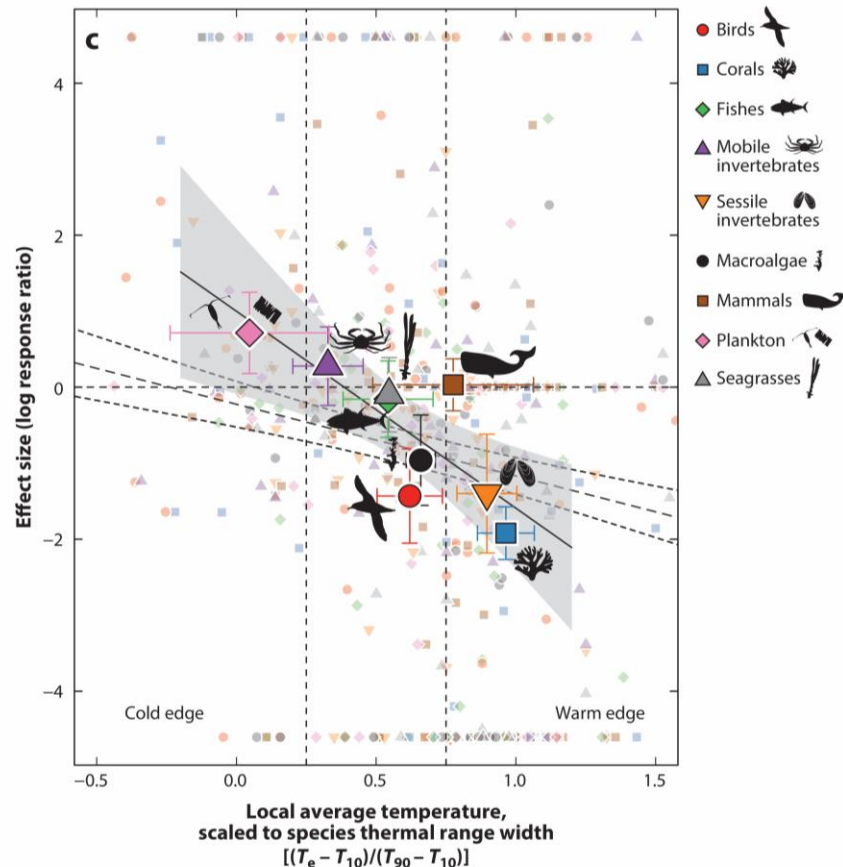
Temperature magnitude exceeds 90th percentile of 30-year historical time series – describes what is extreme for a given location (Hobday et al. 2016, 2018).

But what is extreme for organisms?

- Can use measured or modeled body temperature to approximate a *biological heatwave* (Ma et al. 2025, Noble et al. 2025)
 - ...but still ignores *how* organisms experience heat.



SESSILE INVERTEBRATES EXPERIENCE LARGEST NEGATIVE EFFECTS



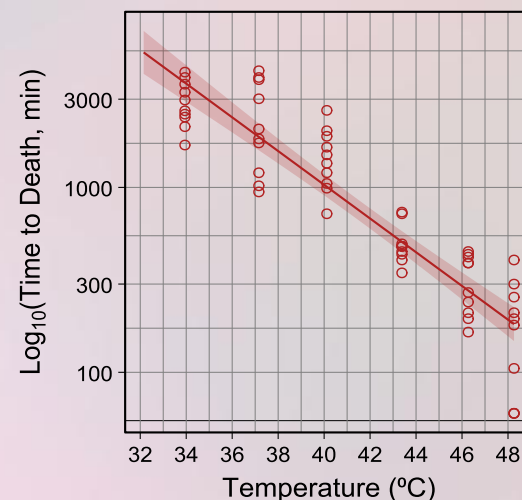
THOMAS NAGEL



What Is It Like
to Be a ~~Bat?~~
Really Hot
Invertebrate?

A BIOLOGICAL APPROACH: THERMAL TOLERANCE LANDSCAPES

- Important to consider exposure **duration** when measuring thermal tolerance - temperature as a dose.
- Time to death follows a log-linear pattern across taxa (Rezende et al. 2014).
- Creates a **continuous landscape** of survival probability across *acute* and *chronic* timescales.

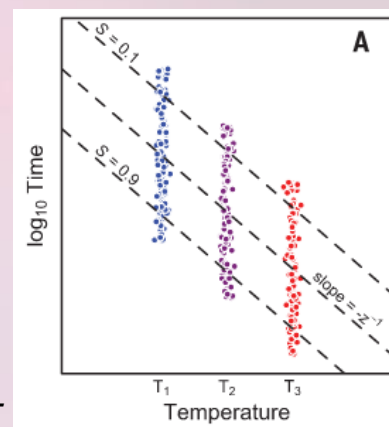


$$\log_{10} \text{time} = \frac{CT_{max} - Temp}{z}$$

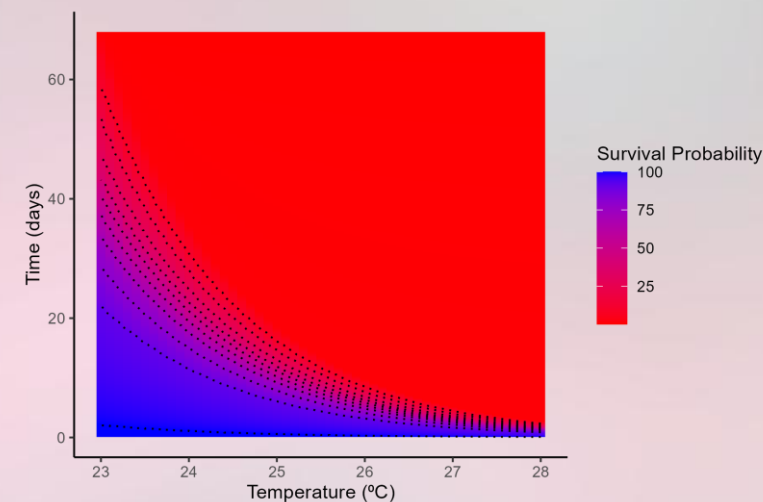
CT_{max} Intercept, when $\log_{10}$ (0 minute) ~ 0

z Negative inverse of slope, scaling factor of tolerance

$$S_{T_1}(\tau) = S_{T_2} \left(10^{-\frac{1}{z}(T_2 - T_1)} \right) \tau$$

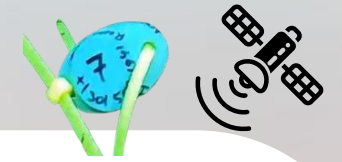


Rezende et al. 2020
Science

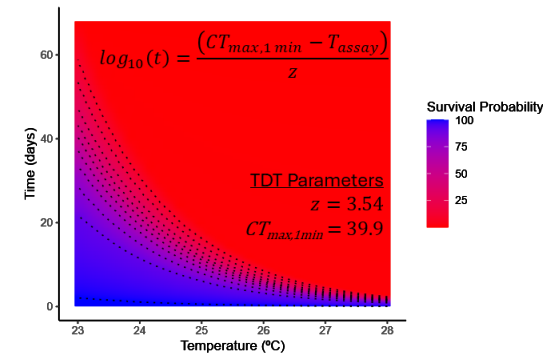


A BIOLOGICAL APPROACH: THERMAL TOLERANCE LANDSCAPES

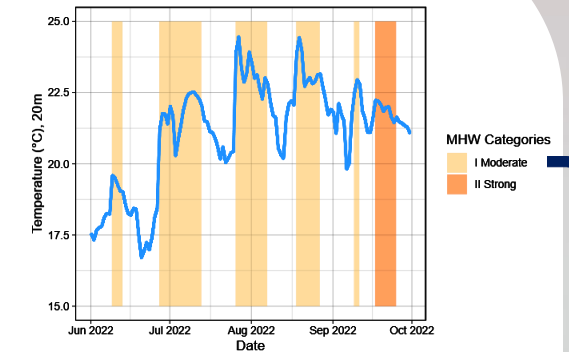
- Calculate **injury accumulation** using dynamic survival models (Rezende et al. 2020, Jørgensen et al. 2021, Ørsted et al. 2024).
 - Inputs: TTL and field body temperature
- For a given heatwave, we can calculate end survival and assess severity.



Thermal Tolerance Landscape

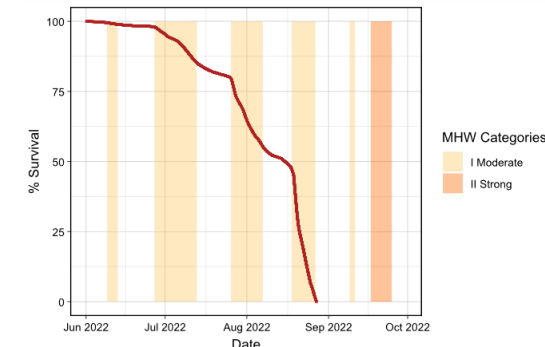


Field Body Temperature



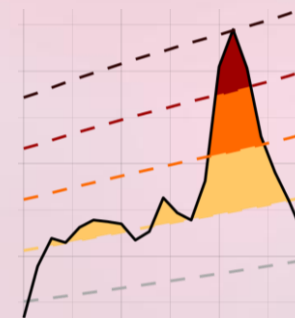
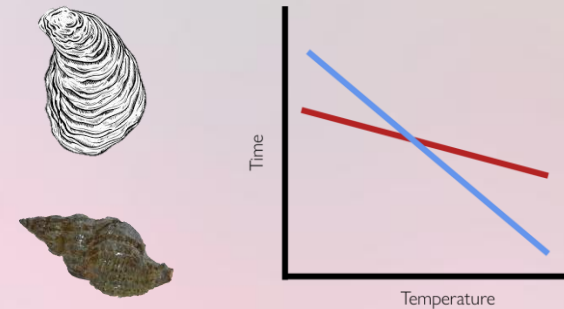
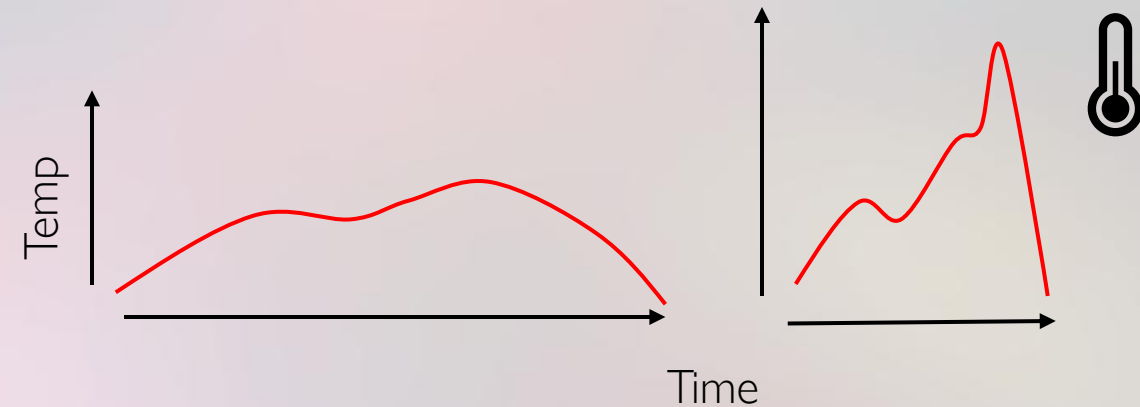
Dynamic Tolerance Algorithm (Rezende et al. 2020, Rezende and Carter 2024)

Survival Prediction

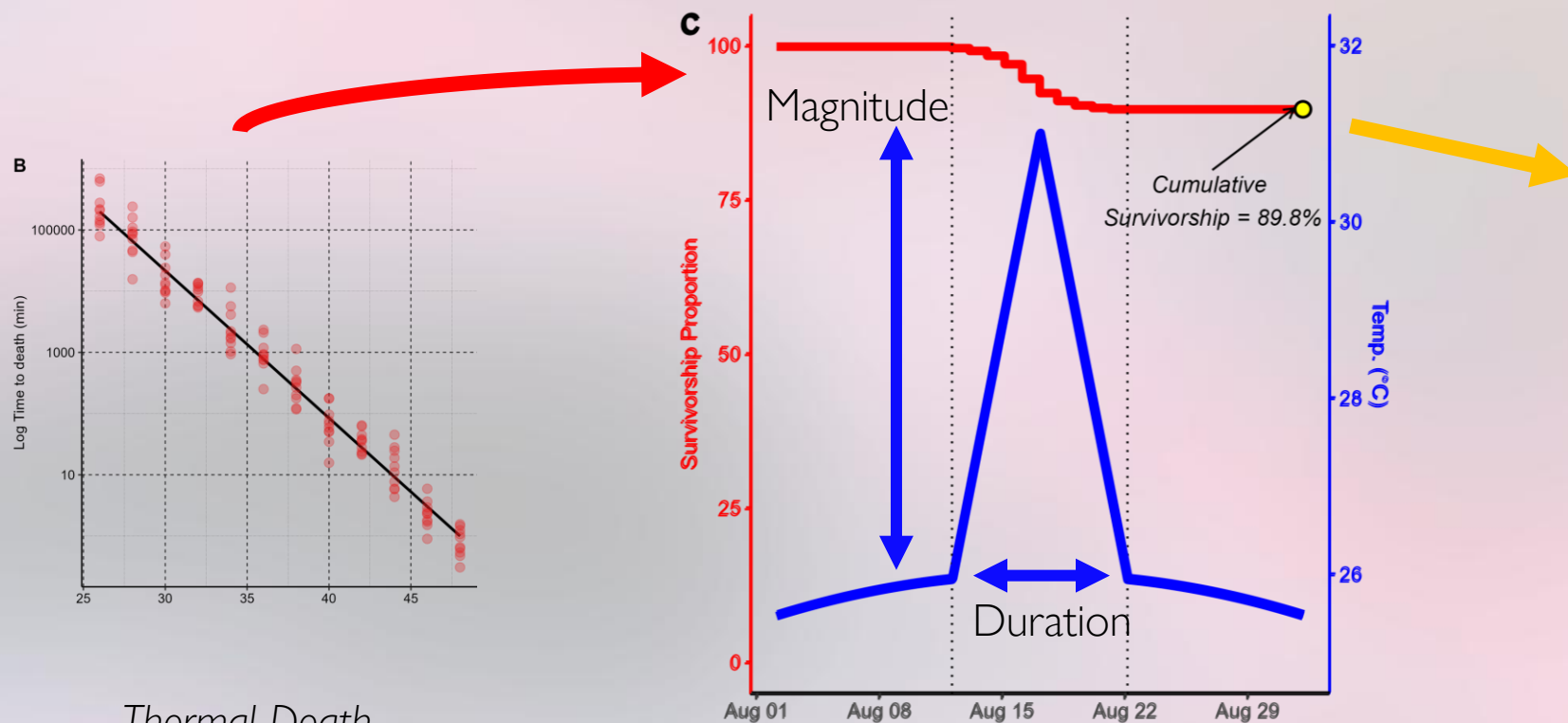


WHAT TYPES OF MHWs ARE THE MOST DETRIMENTAL?

- Multiple candidate metrics:
 - Magnitude (Intensity)
 - Duration
 - Return time
 - Frequency
 - Ramp rate
- How to generalize MHW impacts?
- How do different TDT shapes affect sensitivity to MHWs?
 - Species with different sensitivities
- Do statistical MHWs capture severity of mortality?



BUILDING MHWS AND MASS MORTALITY EVENTS ON YOUR LAPTOP



Replicate 0-8°C × 0-30 days

MHW Magnitude

1.5°C, 1 hour	1.5°C, 1 day	1.5°C, 2 day
1°C, 1 hour	1°C, 1 day	1°C, 2 day
0.5°C, 1 hour	0.5°C, 1 day	0.5°C, 2 day

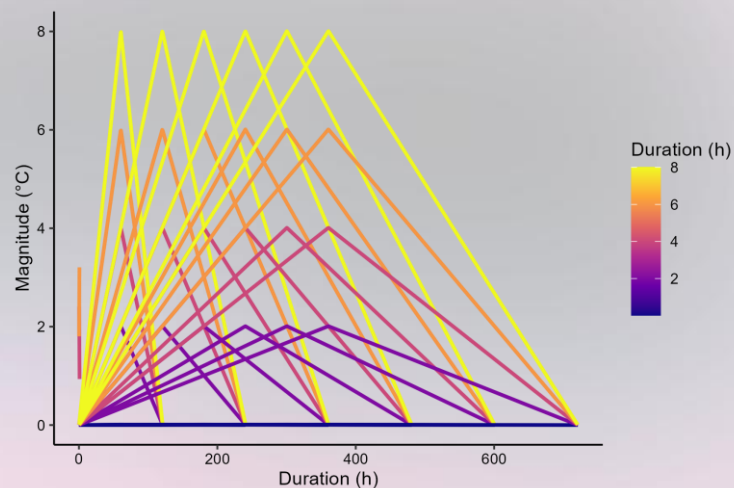
MHW Duration

527 MHW Simulations

Thermal Death
Time Curve

BUILDING MHWS AND MASS MORTALITY EVENTS ON YOUR LAPTOP

De Olde Bestiary of Heatwaves



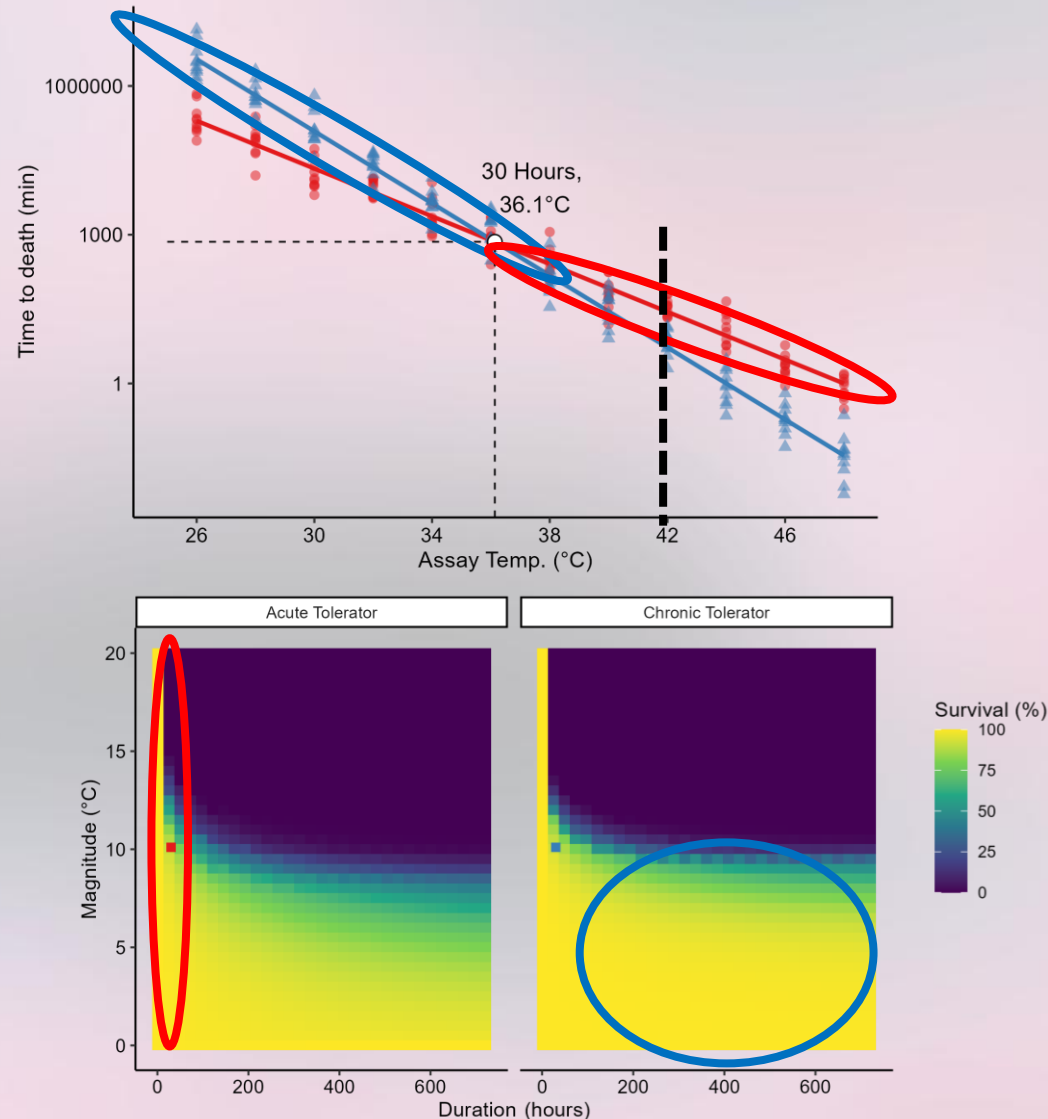
Replicate 0-8°C × 0-30 days

MHW Magnitude	1.5°C, 1 hour	1.5°C, 1 day	1.5°C, 2 day
	1°C, 1 hour	1°C, 1 day	1°C, 2 day
	0.5°C, 1 hour	0.5°C, 1 day	0.5°C, 2 day

MHW Duration

527 MHW Simulations

WINNERS AND LOSERS BETWEEN ACUTE AND CHRONIC HEATWAVE EVENTS

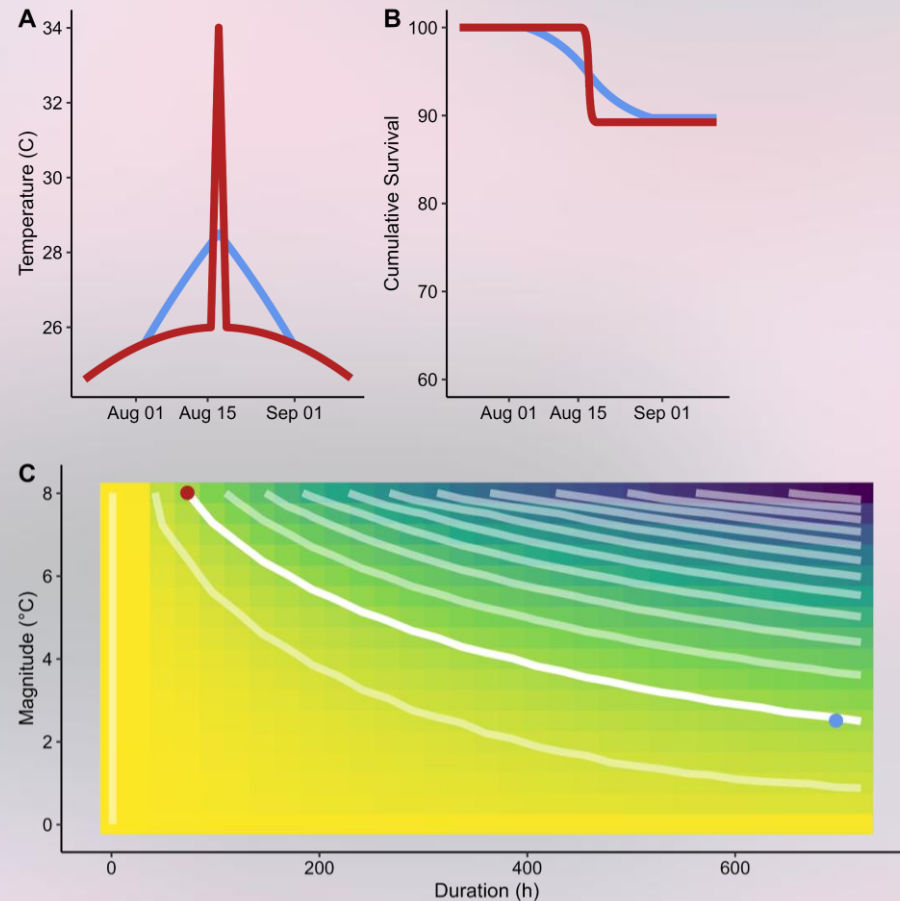


- Two hypothetical species, based on similar values in marine bivalves
- Acute tolerator would be considered more tolerant based on acute thermal tolerance assays
- But beyond 30 hours of exposure, no longer the highest survival probability
- Chronic tolerator performs best during canonical Marine Heatwaves (>5 days)



Villeneuve and White 2024, Journal of Animal Ecology

MAPPING HEATWAVE SIMULATIONS REVEALS SURVIVAL ISOCLINES

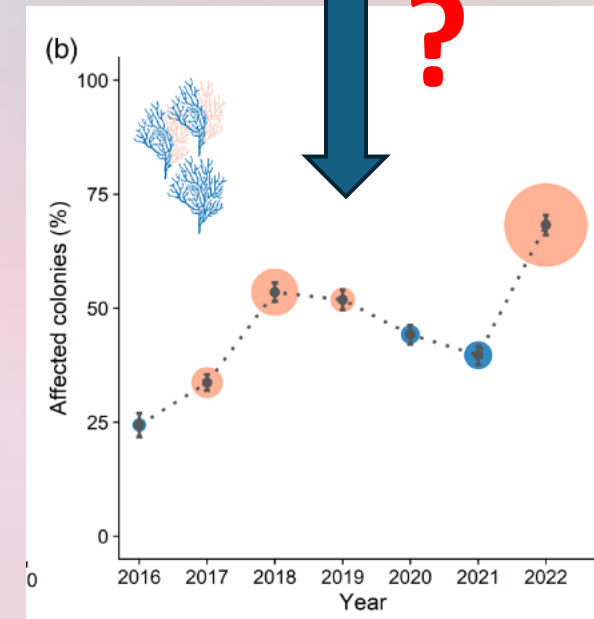
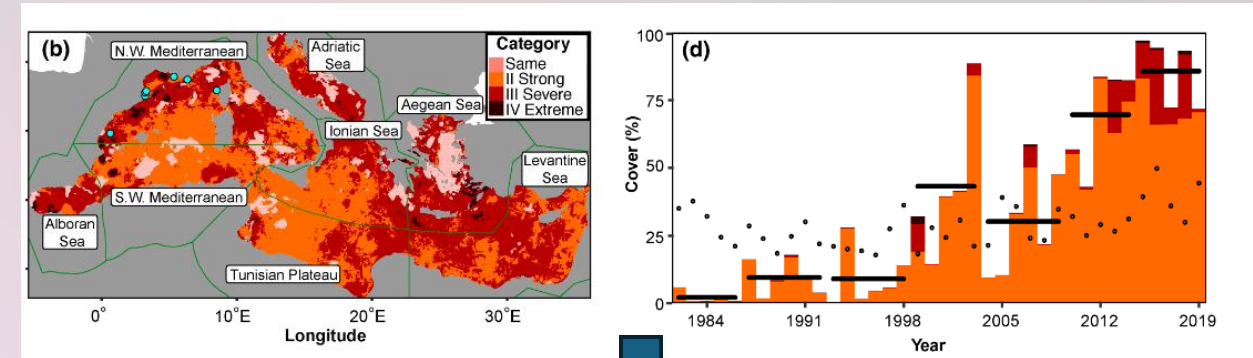
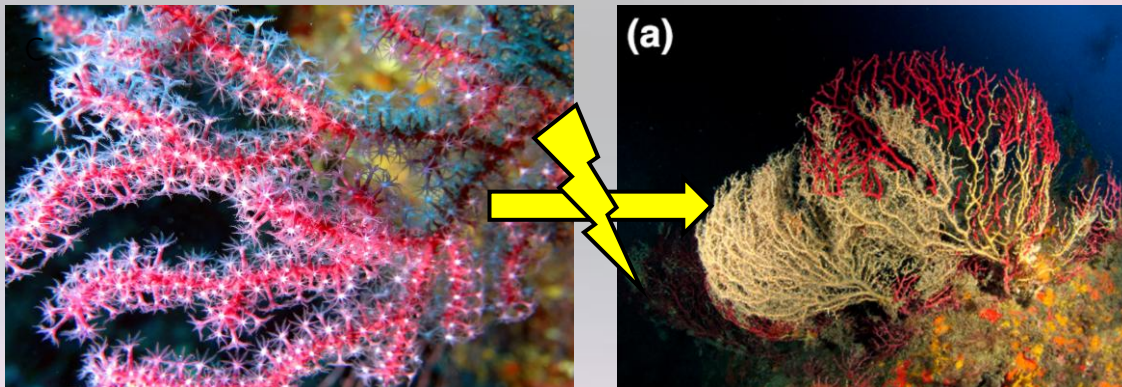


- By tracing isoclines of survival on survival landscapes, we find events that look very different can result in the same survival probability.

So how common are each type of event
in situ?

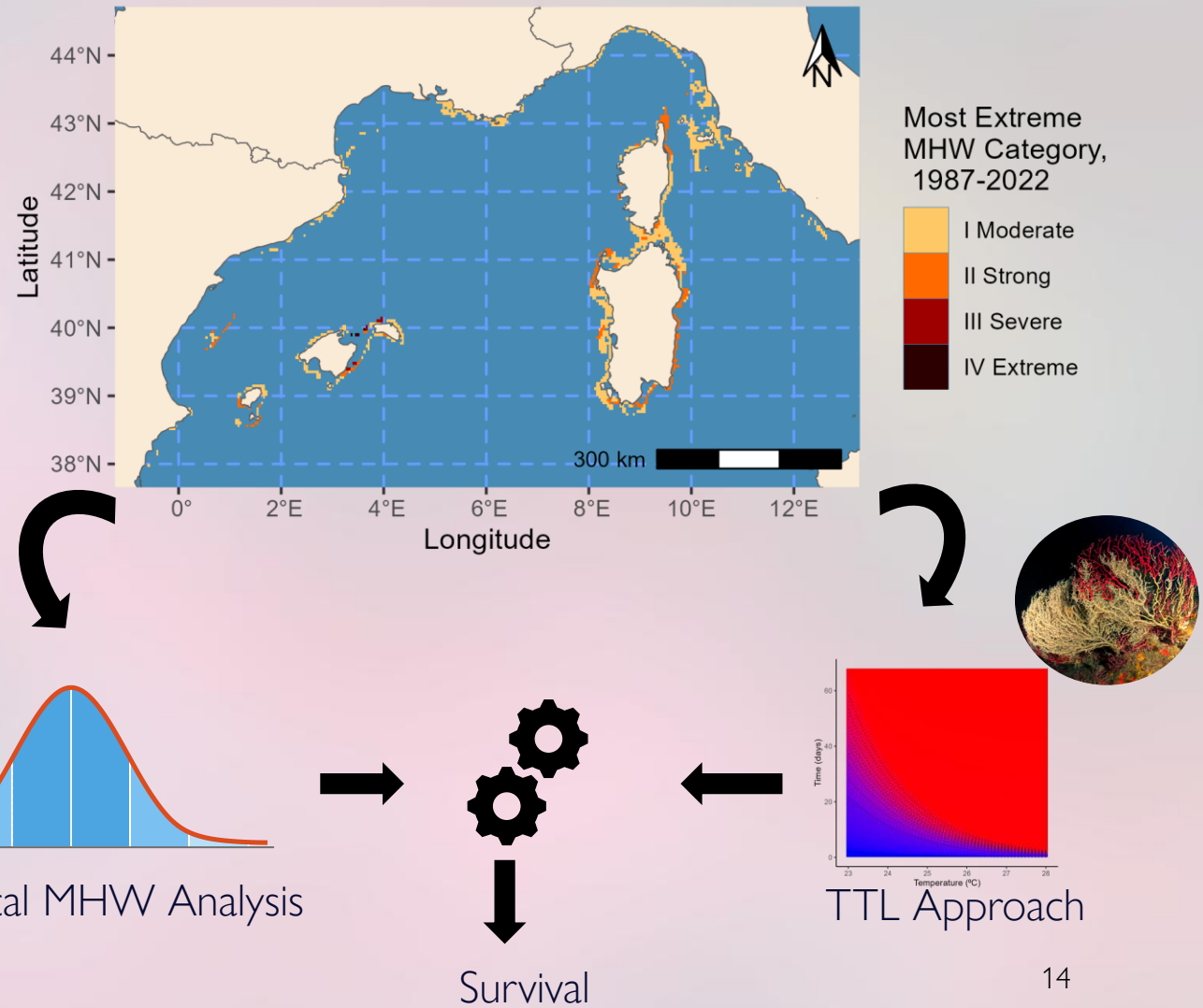
MAPPING BASIN-WIDE MORTALITY OF THE GORGONIAN *PARAMURICEA CLAVATA*

- Intense marine heatwaves in 1999, 2003, and 2022 - extensive mass mortality.
- What types of MHWs are the most common? Which are the most “risky” based on occurrence probability and mortality risk?

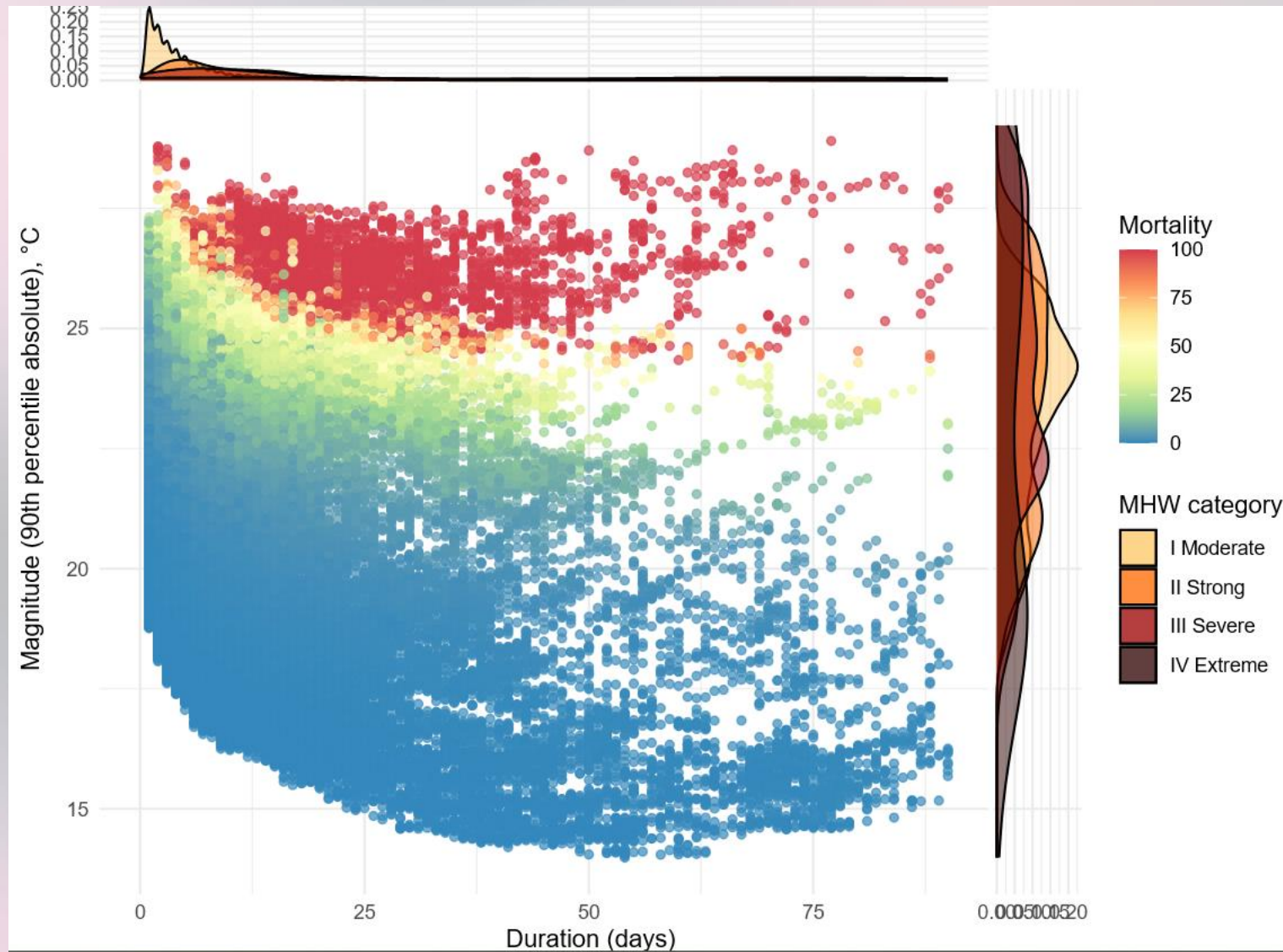


MAPPING BASIN-WIDE MORTALITY OF THE GORGONIAN *PARAMURICEA CLAVATA*

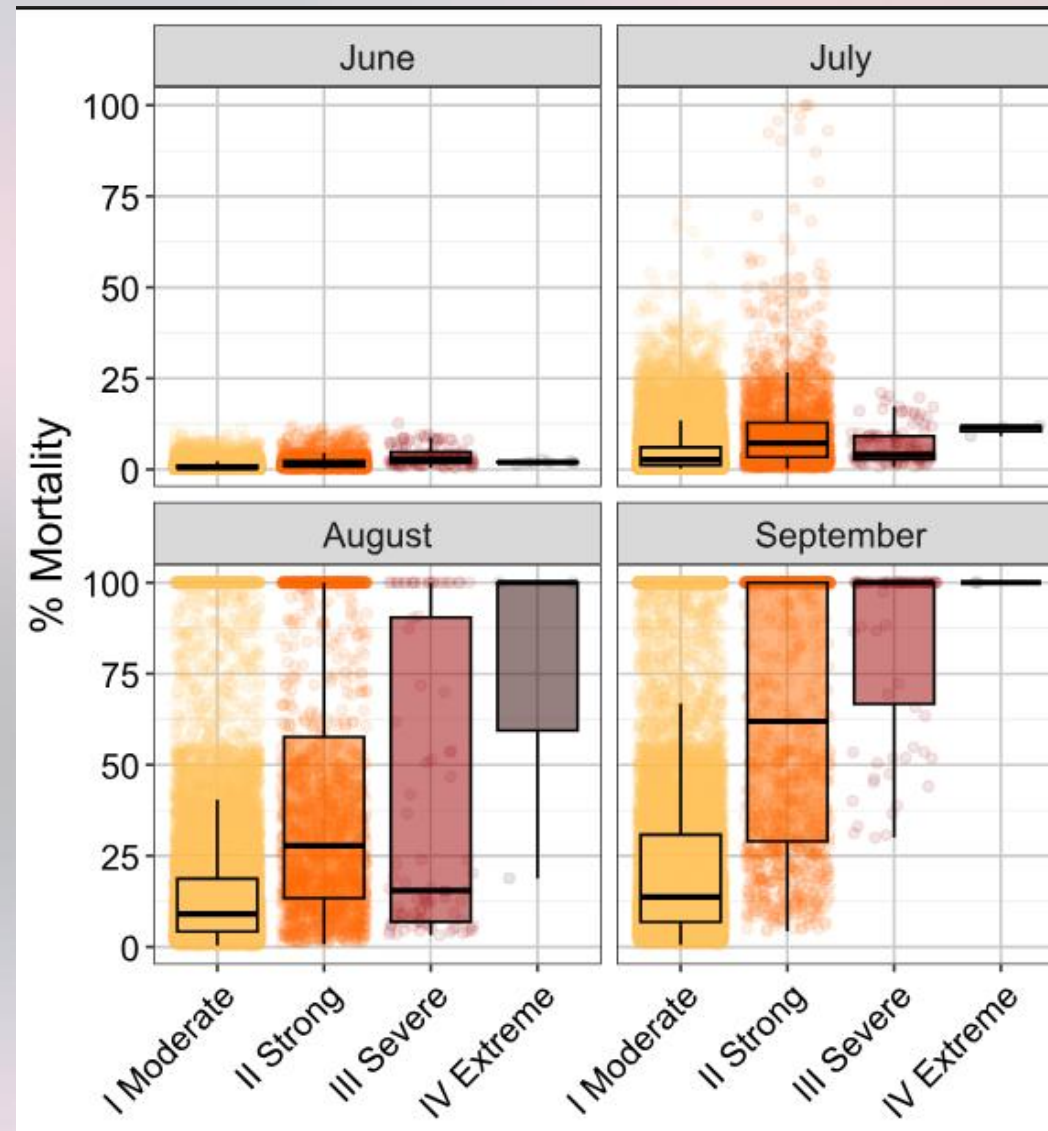
- **Temperature**: Copernicus 4.2km Reanalysis and Forecast ocean temperature (20m depth), clipped to species range.
- **Thermal Tolerance Landscape**: created using data in the literature.
- **Approach**: Identify climatological heatwaves in each range grid cell, then calculate and categorize mortality severity.
 - A mortality estimate over a single heatwave in a single cell.



HEATWAVE IMPACT MAP



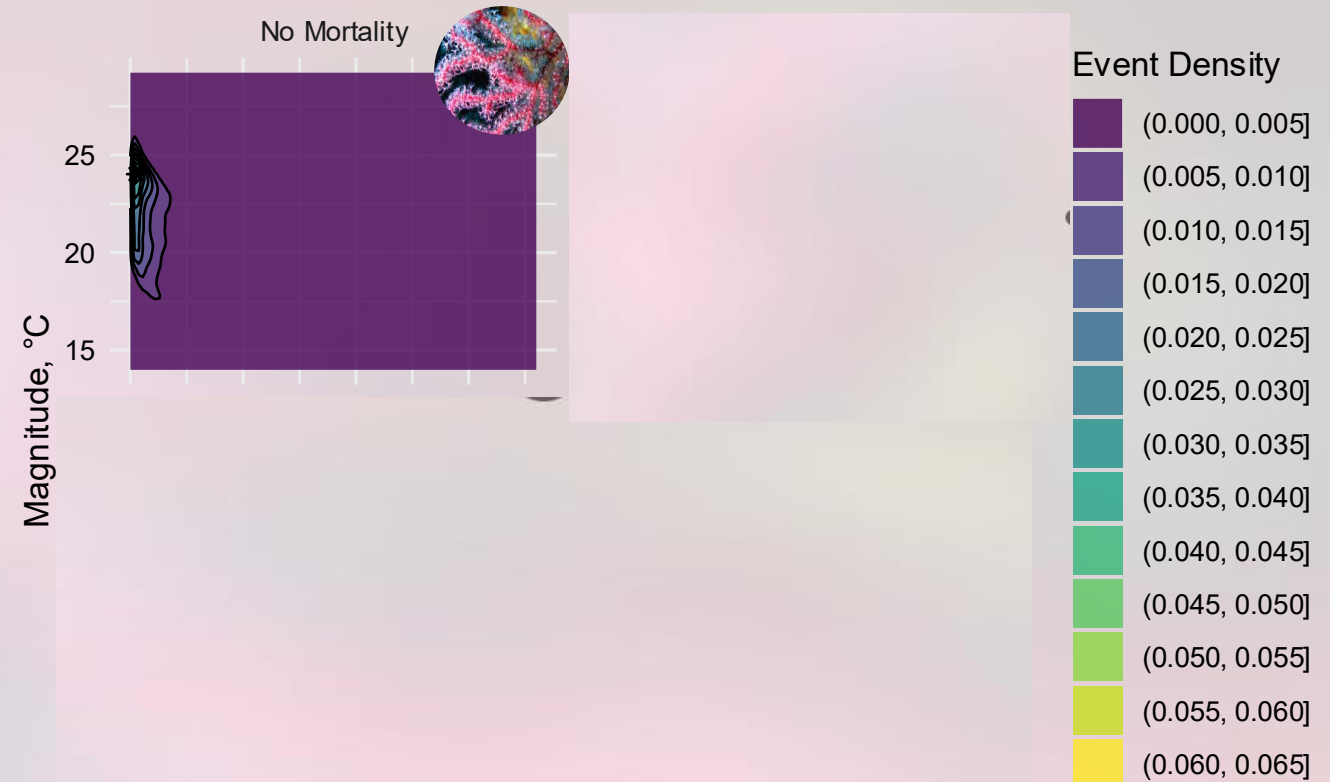
MHW SEVERITY IS NOT ALIGNED WITH MME SEVERITY



DIFFERENT MHW EVENTS CAN HAVE SIMILAR IMPACTS

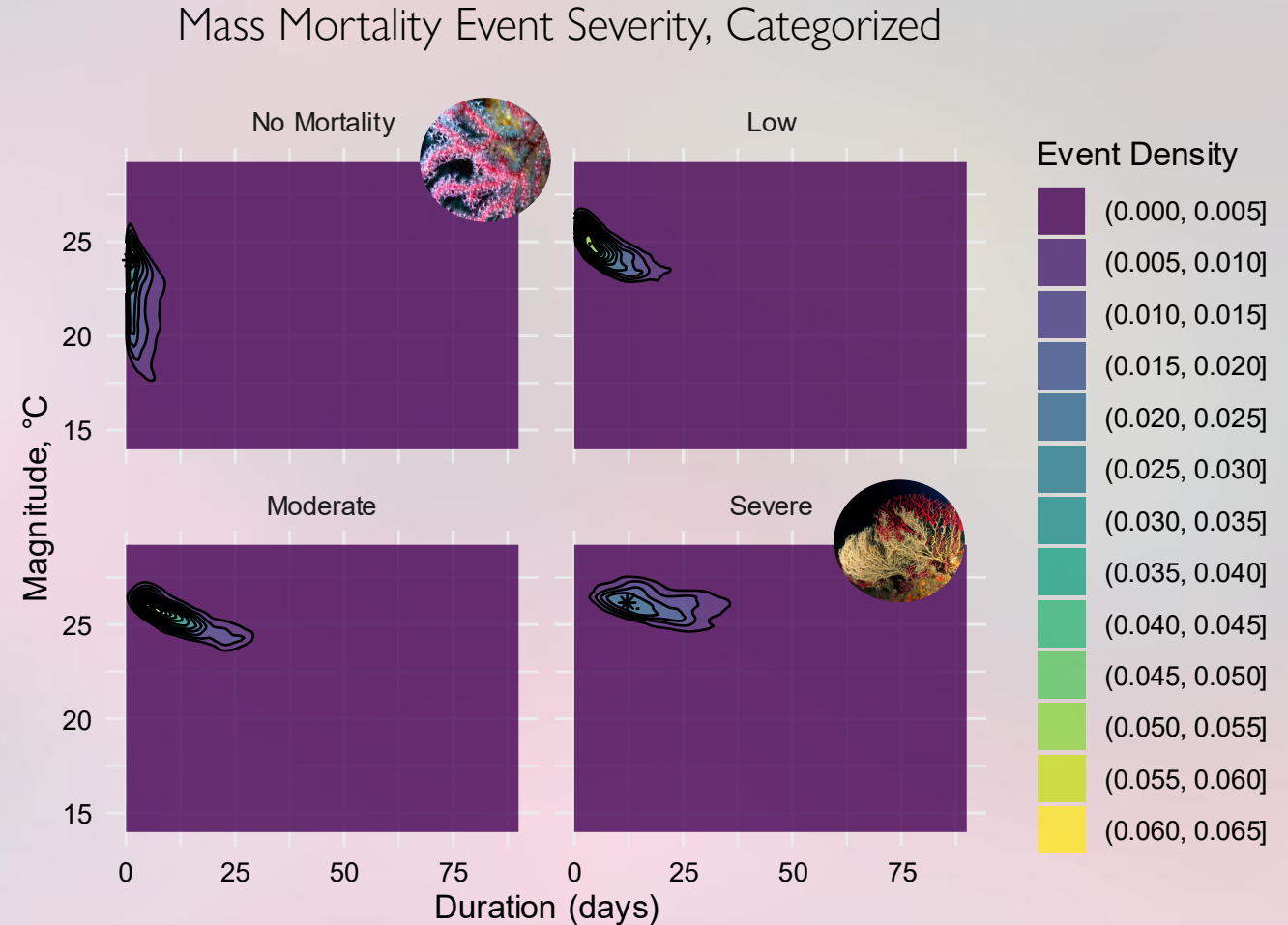
- Survival isoclines exist in modeled survival data – corals with similar mortality can occur over a range of MHW types.

Mass Mortality Event Severity, Categorized



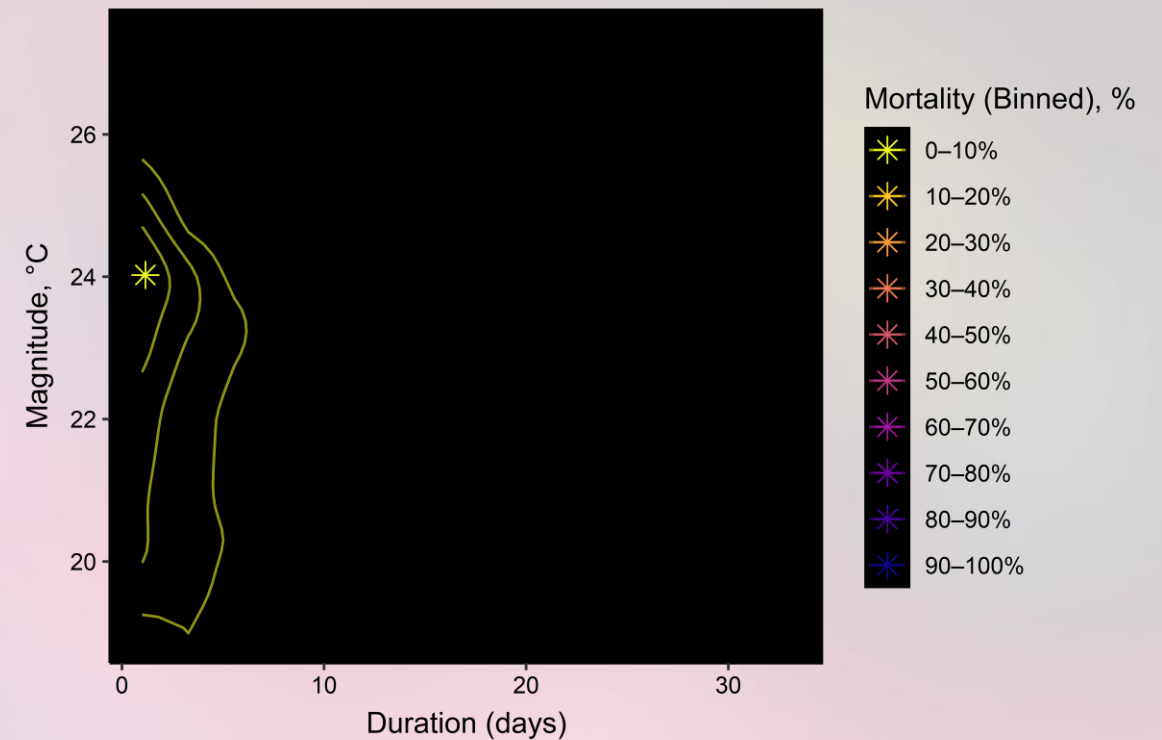
DIFFERENT MHW EVENTS CAN HAVE SIMILAR IMPACTS

- Survival isoclines exist in modeled survival data – corals with similar mortality can occur over a range of MHW types.
- Despite complex MHW profiles, magnitude and duration are major drivers of MHW impact
 - “Ghost” of Thermal Tolerance Landscapes

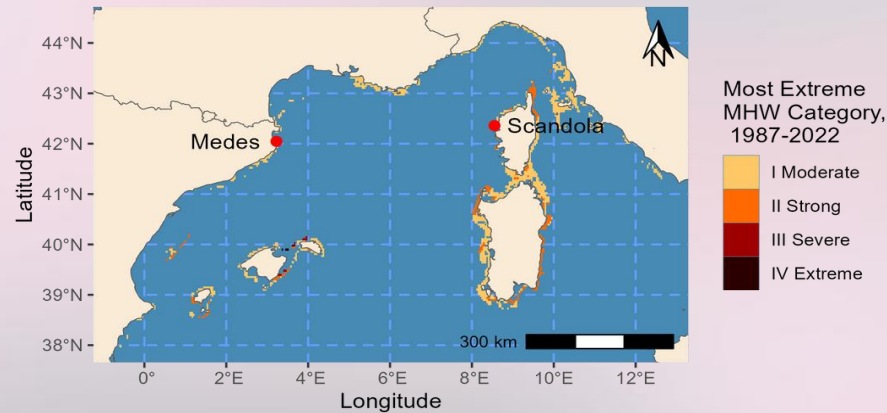


EVENT MAGNITUDE AND DURATION TRADEOFF IMPACT

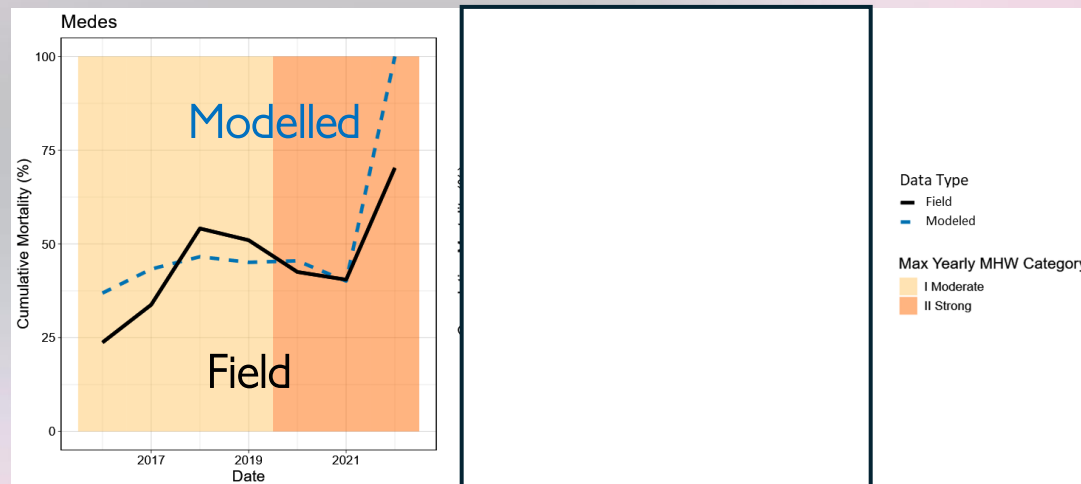
- Magnitude is an important driver of mortality to 30%, after which duration is more important.



PREDICTIVE ACCURACY VARIES BETWEEN SITES

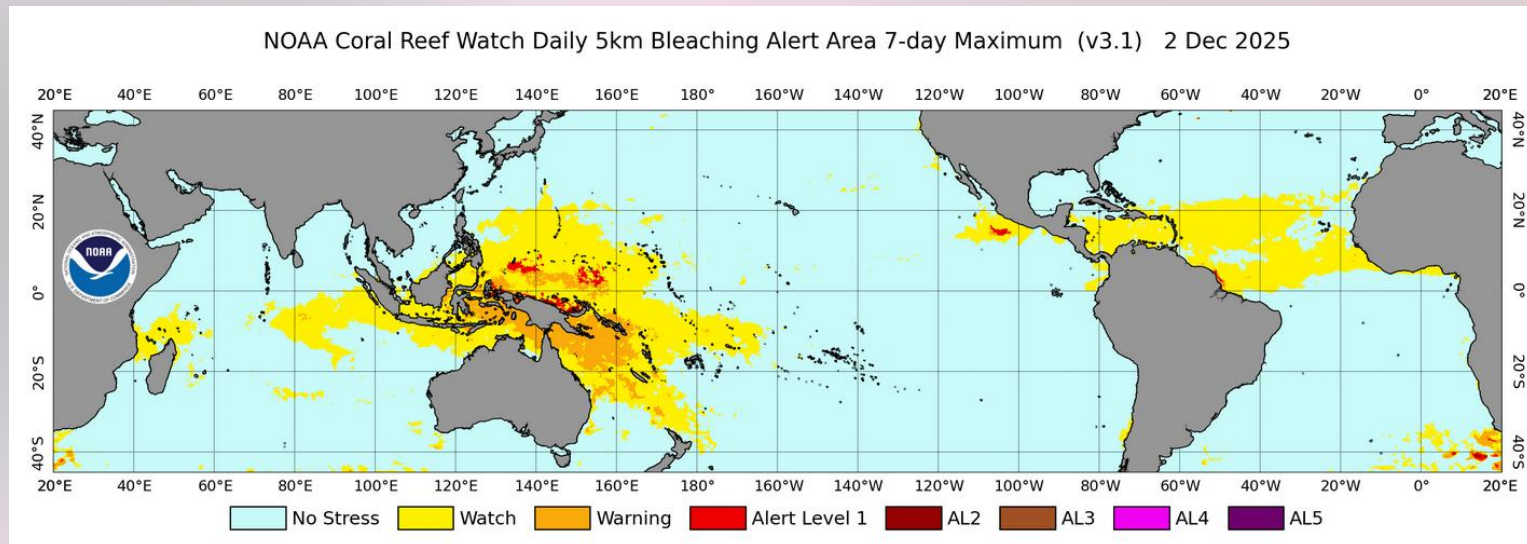


- Field mortality observed at the Medes Islands, ES, and Scandola MPA, FR and compared against modeled mortality
- Good fit of modeled and observed survival at Medes. Local adaptation may be responsible for overprediction at Scandola.
- Demonstrates inaccuracies in a climatological approach to MHW impacts



TOWARDS A HEATWAVE MORTALITY FORECASTING DASHBOARD

- Copernicus reanalysis products forecast 9 days in advance, and recently *hourly* resolution across 141 depths
- Mechanistic models require precise parameterization and are computationally expensive
- Future steps: Statistical emulation of non-closed-form functions and integrating spatial and temporal variation in TTLs



Environment

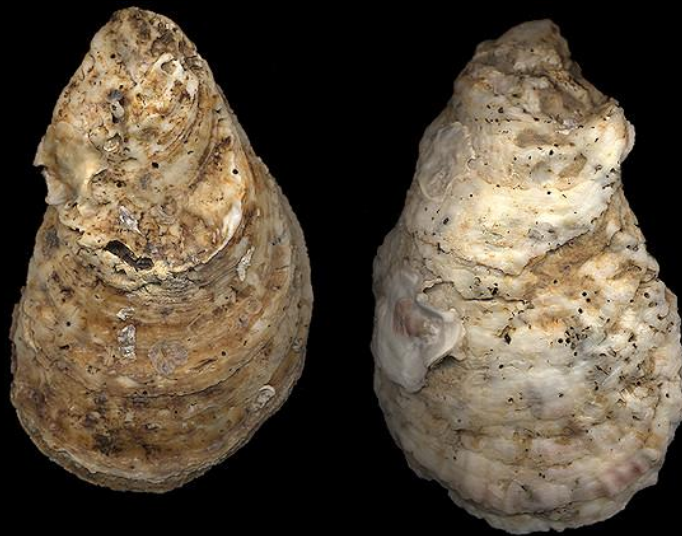


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**University of
New Hampshire**
NH Agricultural
Experiment Station



**University of
New Hampshire**
School of Marine Science
and Ocean Engineering

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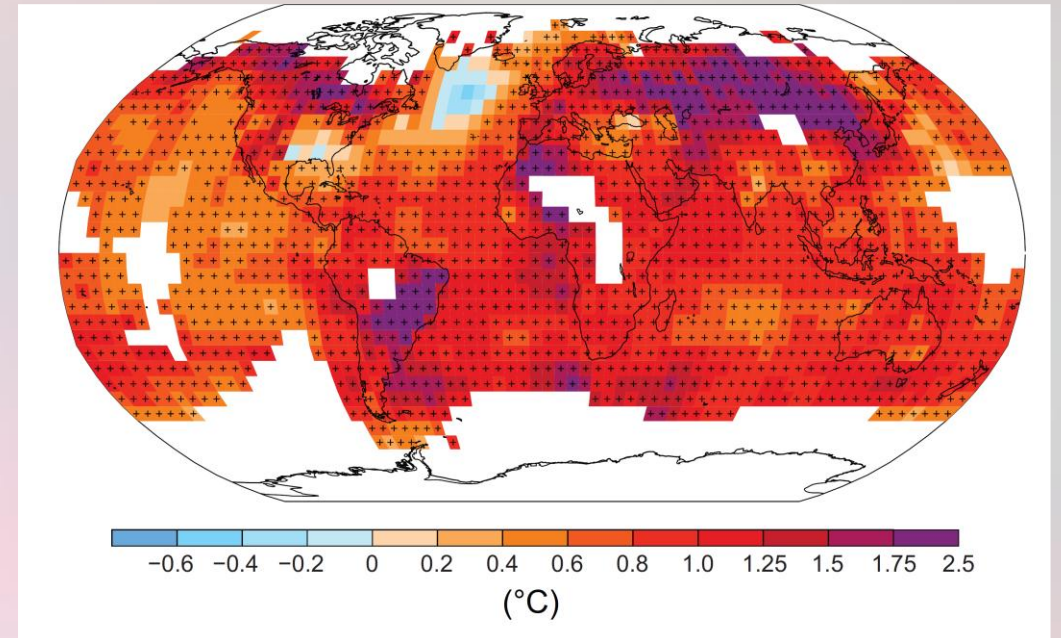
drew.villeneuve@unh.edu

Slides

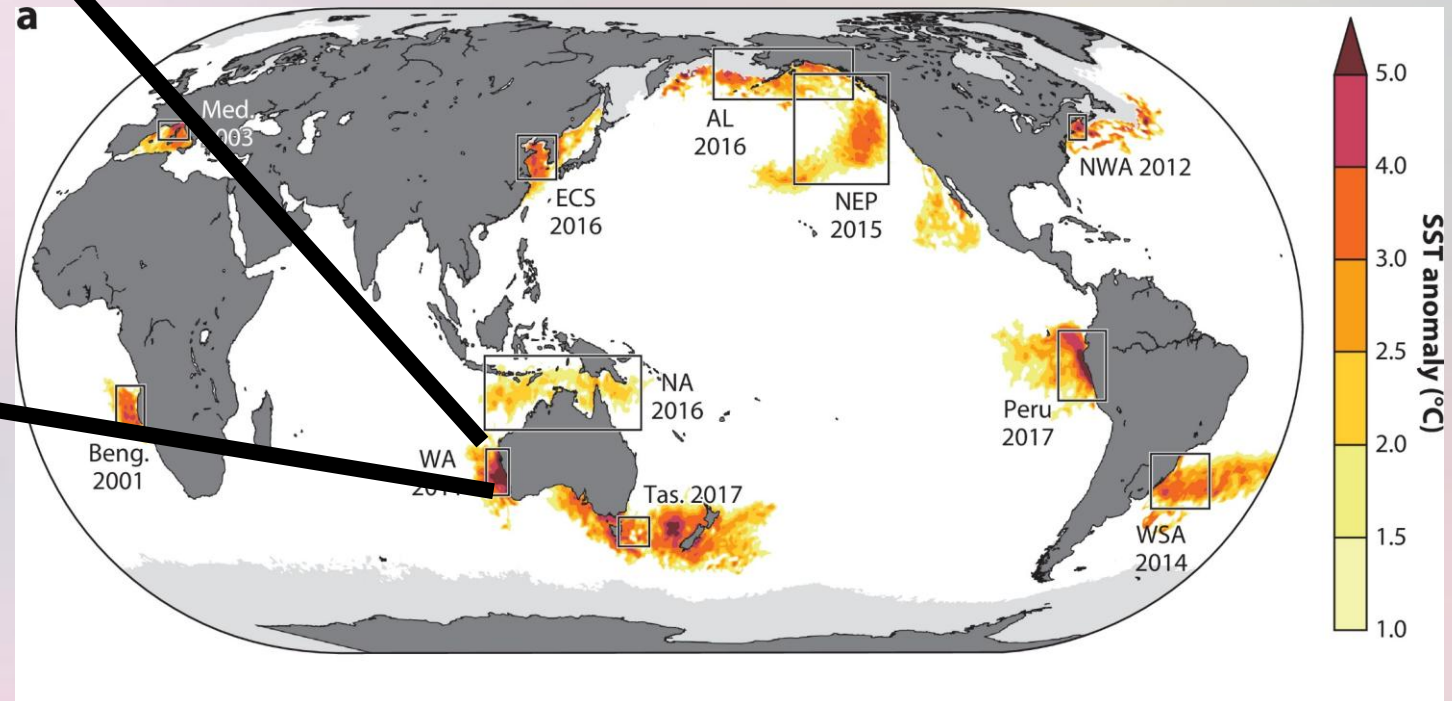
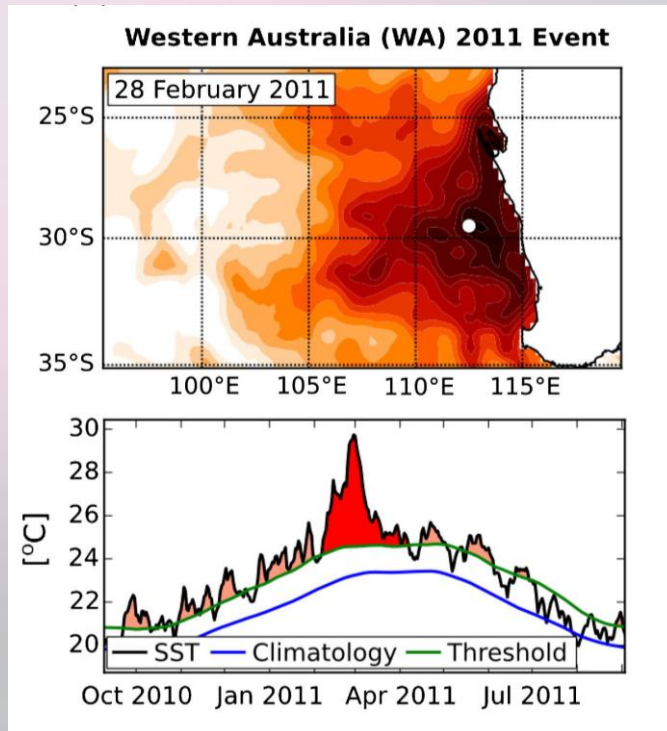


HOW TO FORECAST GLOBAL CHANGE IN THE WORLD'S OCEANS?

- Nonstationary, novel climatic system
- How to forecast organismal and ecosystem responses *mechanistically* to climate change?
 - Temporal variation
 - Geographic variation
 - Extreme events
- Across scales
- Applications for species and ecosystem recovery, socioecological linkages



MARINE HEATWAVES (MHWS) – DEPARTURES FROM CLIMATOLOGICAL NORMS



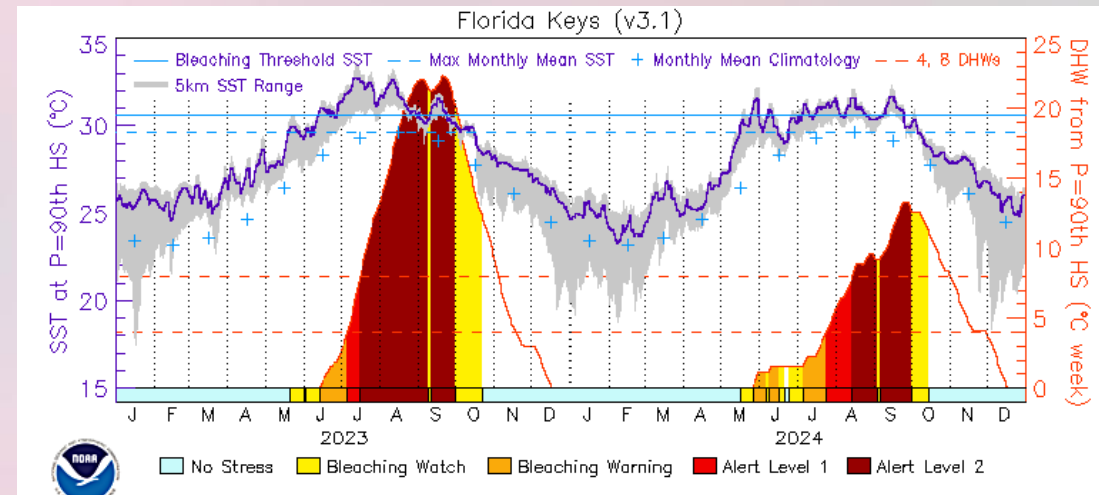
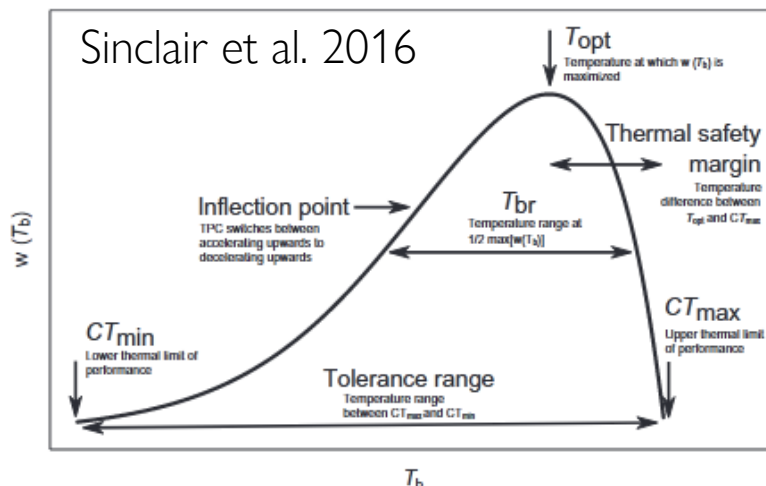
HOW TO PREDICT HEATWAVE MASS MORTALITY EVENTS?

Threshold

- Temperature at which biological rates cease
 - LT_{50} or CT_{max}
- Sensitive to methodology
 - Ramping rate, time of exposure
- Binary state – alive or dead

Degree Heating Weeks (DHWs)

- Accounts for time spent at stressful temperatures
- Includes time dimension, but based on climatological thresholds





Mass mortality event



Marine mammals/birds



Wild fish/shellfish



Bivalve shellfish



Marine forests

Habitat impacts



Seagrass meadows

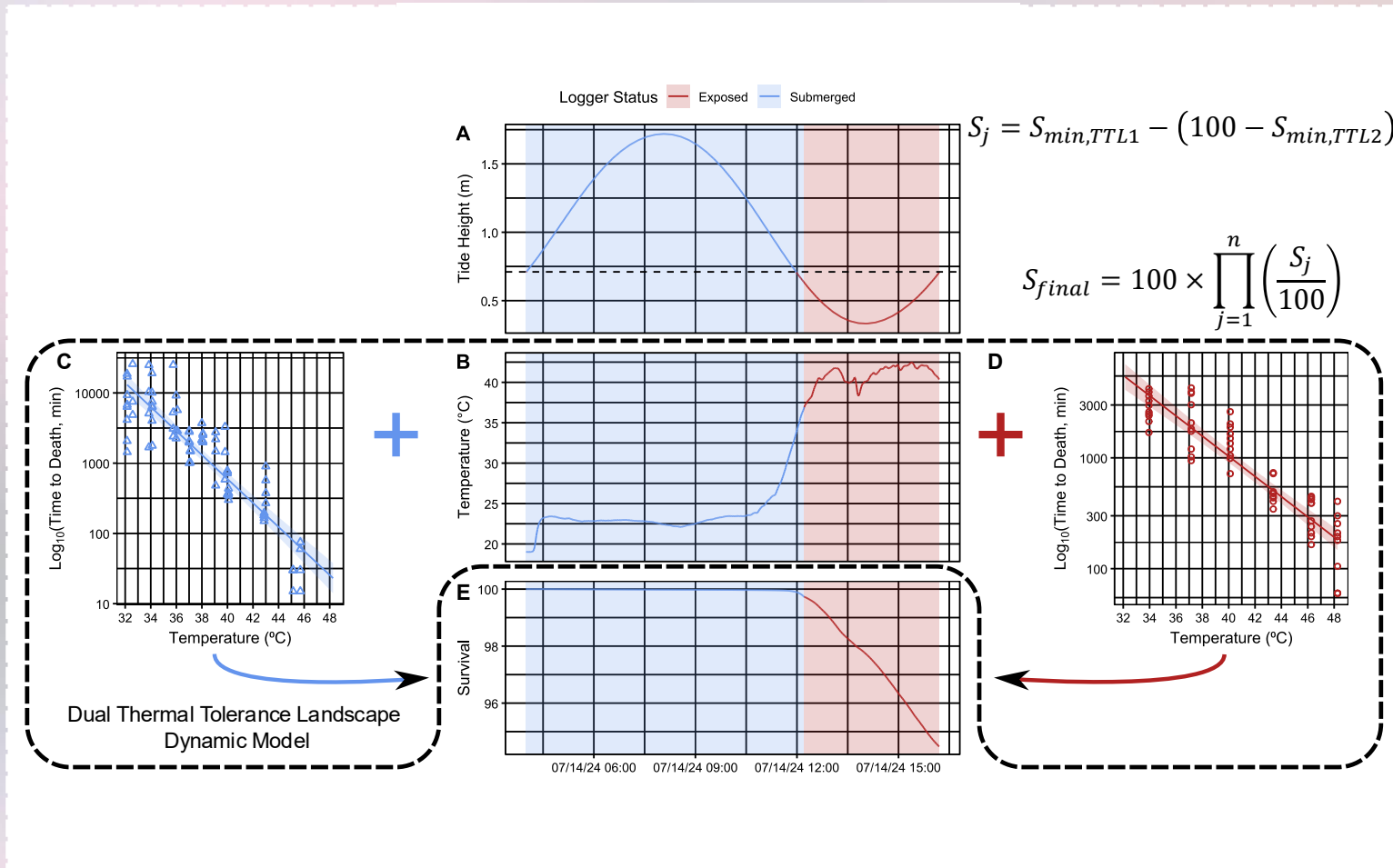


Coral reefs

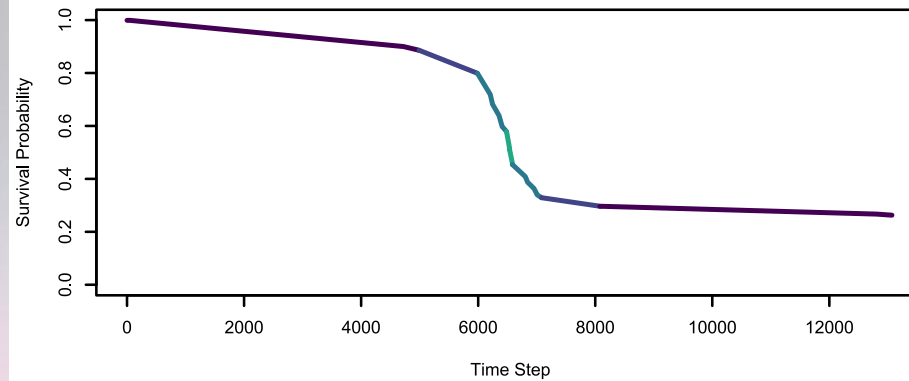
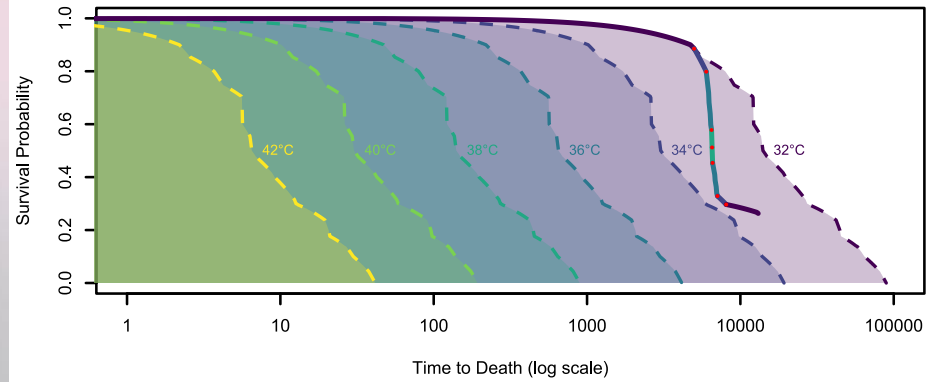
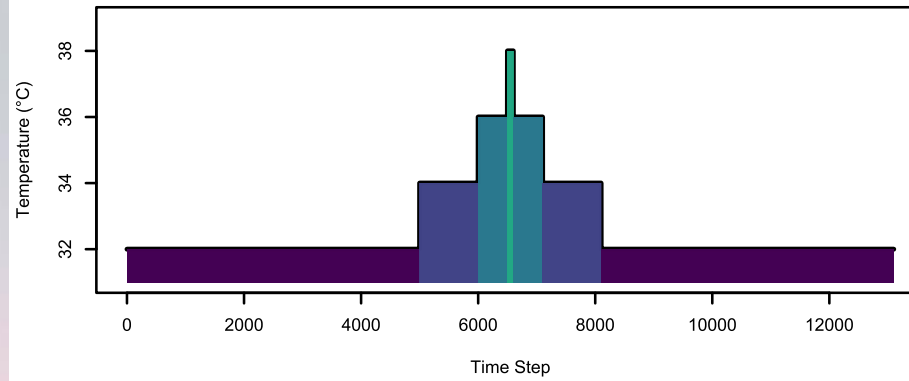


Biogenic reefs (noncoral)

Eastern oyster expresses a similar acute/chronic tradeoff depending on intertidal exposure



- Stress begins to accumulate at the beginning of each tide cycle (defined as immersion-emersion)
- Upon emersion, stress accumulated during high tide is included in the calculation of low tide stress
- Re-immersion in water at high tide resets stress



$$\Delta S_{T(\tau)} \approx \left. \frac{dS(\tau)}{d\tau} \right|_{T(\tau)} \Delta \tau$$

CAN WE PREDICT HEAT-INDUCED MASS MORTALITY EVENTS?



- Predicting oyster mortality in Great Bay, NH using thermal tolerance landscapes
- Re-assessing climatological heatwaves with TDT curves
- Application towards predicting gorgonian mortality in the Western Mediterranean and mismatches with heatwave intensity

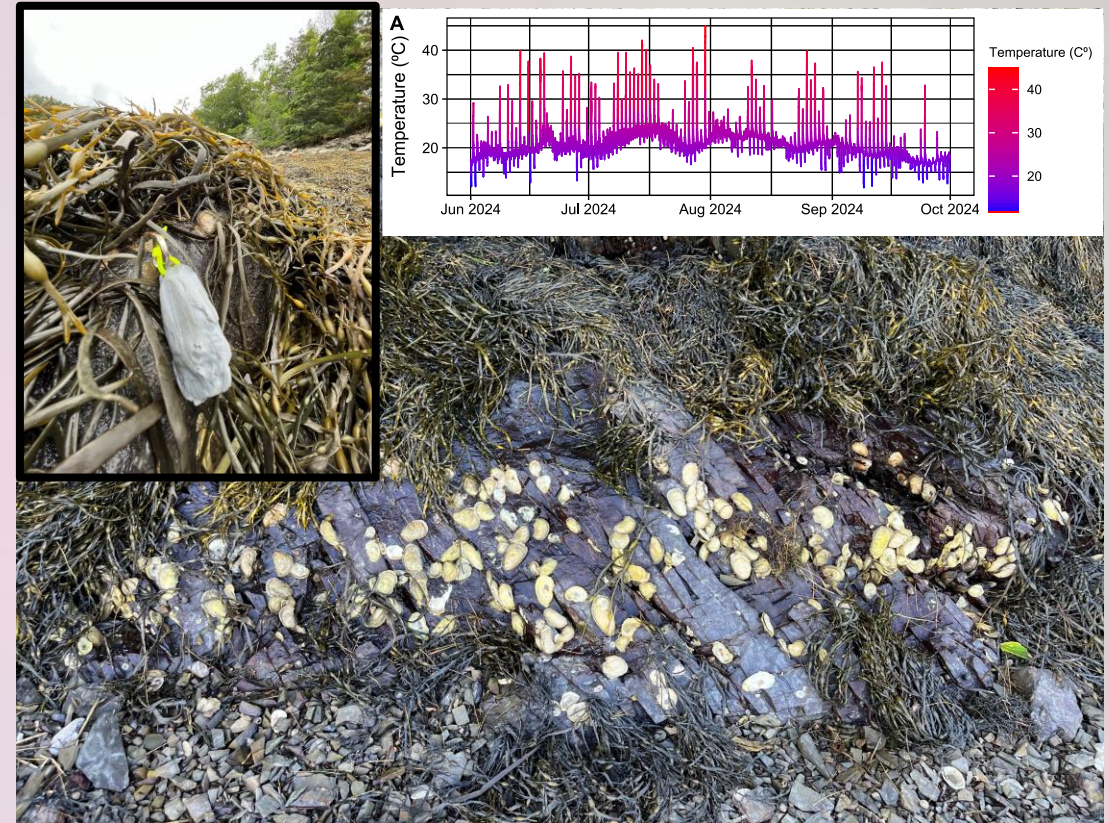


Forecasting Goals: Predict...

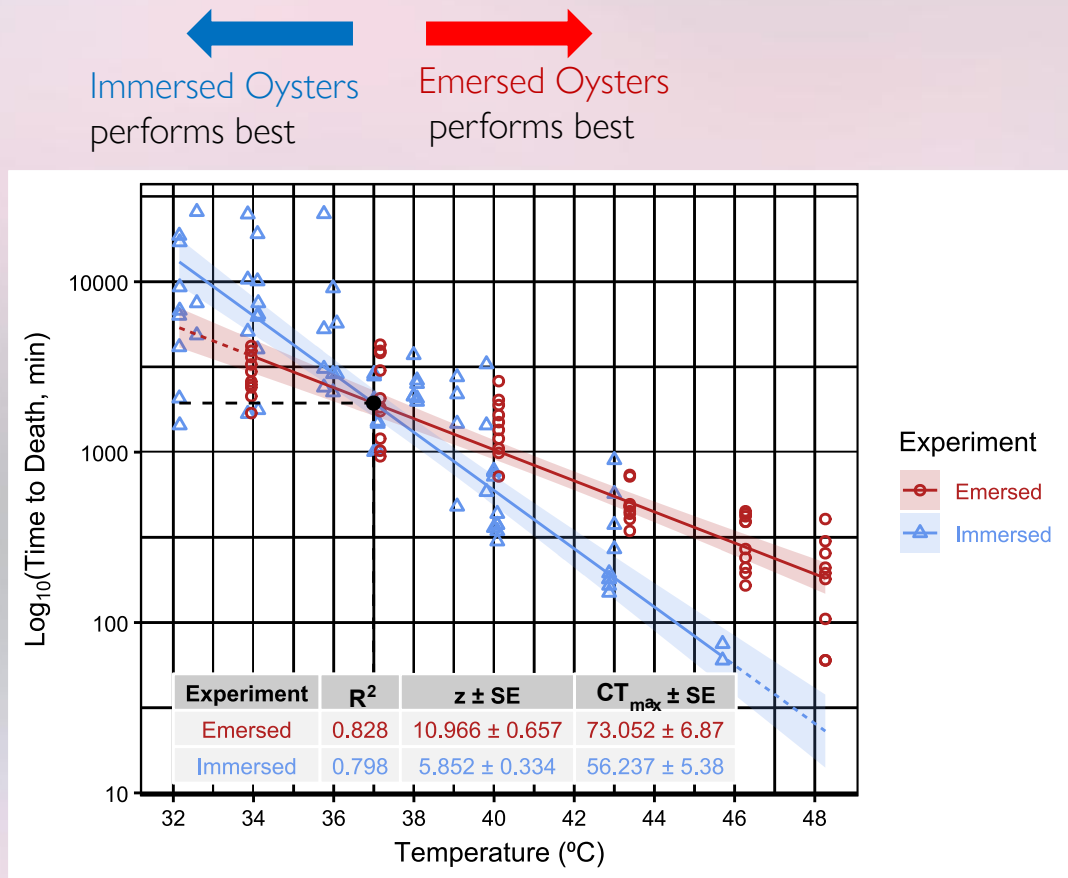
- 1) When MHW conditions cause mass mortality events
- 2) What the extent of damage is resulting from these MHW events

HOW TO PREDICT MORTALITY IN A SPECIES WITH VARYING TTL?

- Eastern oyster (*Crassostrea virginica*) lives in intertidal areas of Great Bay, NH
 - Does thermal tolerance change with tide exposure?
 - Do eastern oysters experience mortality events during MHWs in New Hampshire?
- Need to:
 - Perform time to death experiments in air and water
 - Construct algorithm to model mortality in an intertidal scenario

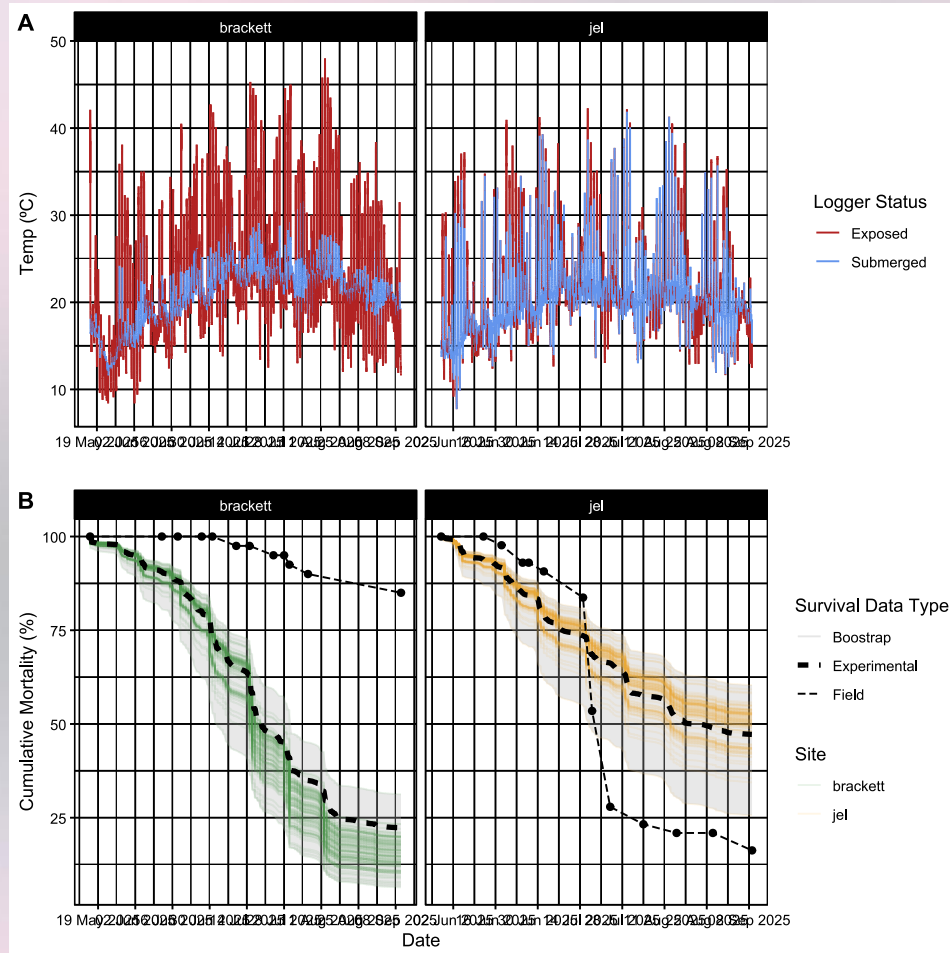


Eastern oyster expresses a similar acute/chronic tradeoff depending on intertidal exposure



- Above 37°C, oysters survive longer in air than water.
- Longer tolerance in air likely due to metabolic depression via anaerobic metabolism pathways (Hui et al. 2020).
- Potential for evaporative cooling
 - Oysters gape under aerial heat stress

Eastern oyster expresses a similar acute/chronic tradeoff depending on intertidal exposure



- Models tend to overpredict mortality – until they don't!
 - Construction, turbidity, disease may be responsible for mortality at JEL
- Heat hardening may be responsible for overperformance of field populations
 - Acclimation was hotter in field surveys than laboratory experiments

Methods | Measure Time to Death



Infrared
Cardiac Sensor



CML Seawater
System

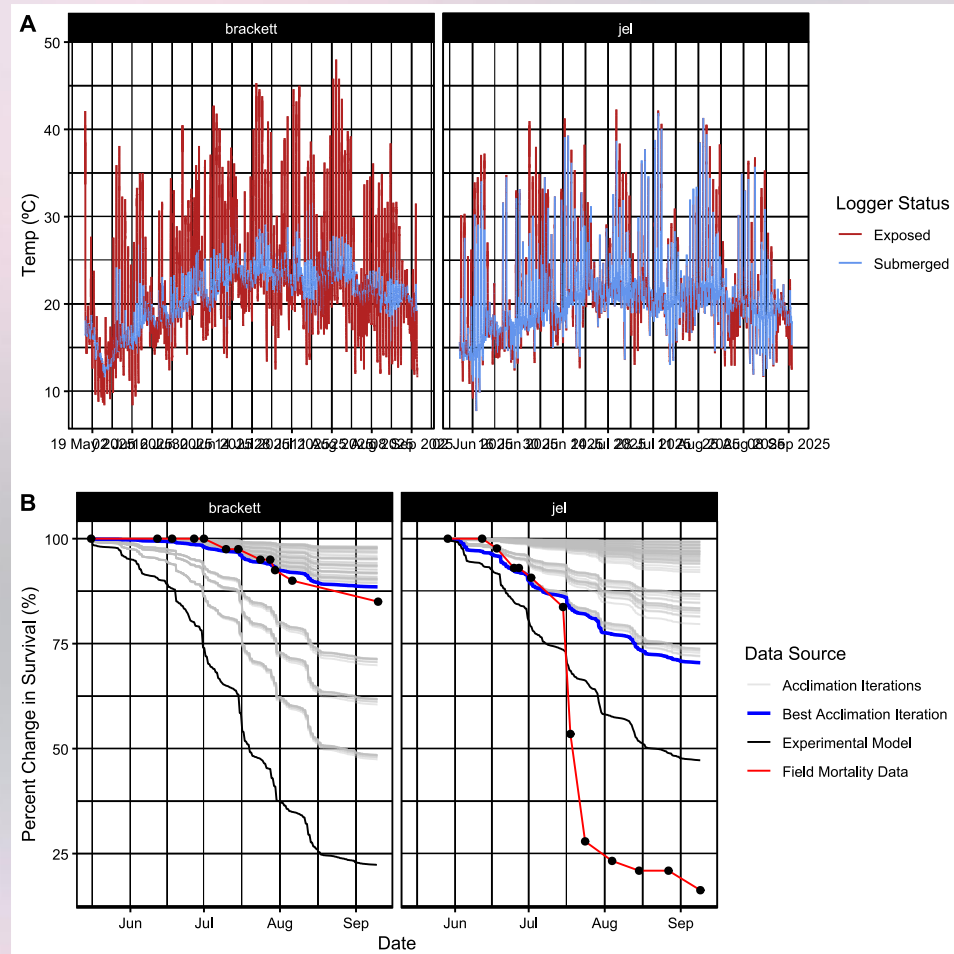


Air Incubators

- Record cardiac activity of individual oysters in submerged (high tide) and exposed (low tide) laboratory assays
- Measure time of death, as indicated by cessation of cardiac activity
- Validated with field mortality

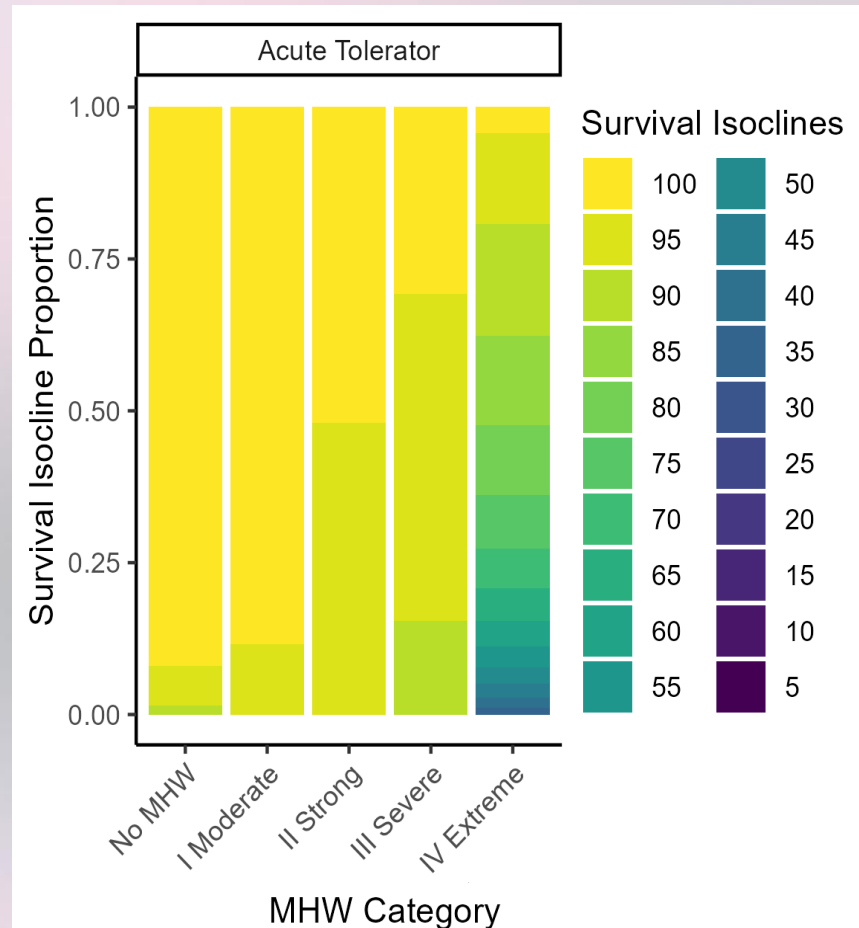


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- Models tend to overpredict mortality – until they don't!
 - Construction, turbidity, disease may be responsible for mortality at JEL
- Heat hardening may be responsible for overperformance of field populations
 - Acclimation was hotter in field surveys than laboratory experiments
- Stress accumulation algorithm appears to work, but TTLs from lab experiments don't match field performance

MHW AND MORTALITY EXTREMITY MISMATCH



- Mass mortality event severity is generally consistent with less severe MHW events
- However, under extreme MHWs, survival can range widely (100%-45%)

Is this observable in a real system?